TOPIC : FIRST FIT

OUTPUT :

```
maitreyee@localhost:~
ab.sh      code       maii.c     os1        Public     sh         Videos
a.out      Desktop    mait       os.c       REC        sjf        while.sh
awk        devi       maithu     os_lab     roundrobin sjf.c      whilw.sh
bank       Documents  maitre     os_lab1    roundrobin.c sphore   XYZ
banker.c   dog        mash       os_lab1.c  salary     student.txt
bankers    Downloads  mithii     os_lab.c   sample.c   table.sh
[maitreyee@localhost ~]$ gcc bestfit.c -o bestfit
[maitreyee@localhost ~]$ ./bestfit
Enter number of memory blocks:
5
Enter size of each block:
100 200 300 400 500
Enter number of processes:
4
Enter size of each process:
112 113 114 115

Process No.     Process Size    Block No.
1               112             2
2               113             3
3               114             3
4               115             4
[maitreyee@localhost ~]$ vi bestfit.c
[maitreyee@localhost ~]$
```

Activities Terminal Apr 7 14:11

maitreyee@localhost:~ — vi bestfit.c

```c
#include <stdio.h>

int main() {
    int m, n;

    // Input number of memory blocks
    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);

    int blockSize[m], remainingSize[m];

    // Input block sizes
    printf("Enter size of each block:\n");
    for (int i = 0; i < m; i++) {
        scanf("%d", &blockSize[i]);
        remainingSize[i] = blockSize[i];
    }

    // Input number of processes
    printf("Enter number of processes:\n");
    scanf("%d", &n);

    int processSize[n], allocation[n];

    // Input process sizes
    printf("Enter size of each process:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &processSize[i]);
        allocation[i] = -1; // Not allocated initially
    }

    // First Fit Allocation
    for (int i = 0; i < n; i++) {
```

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maitreyee@localhost:~ — vi bestfit.c

```c
    int processSize[n], allocation[n];

    // Input process sizes
    printf("Enter size of each process:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &processSize[i]);
        allocation[i] = -1; // Not allocated initially
    }

    // First Fit Allocation
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            if (remainingSize[j] >= processSize[i]) {
                allocation[i] = j;
                remainingSize[j] -= processSize[i];
                break;
            }
        }
    }

    // Output allocation
    printf("\nProcess No.\tProcess Size\tBlock No.\n");
    for (int i = 0; i < n; i++) {
        printf("%d\t\t%d\t\t", i + 1, processSize[i]);
        if (allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }

    return 0;
}
```